



APLAC REQUIREMENTS AND GUIDANCE ON THE ACCREDITATION OF A REFERENCE MATERIAL PRODUCER

PURPOSE

The requirements and guidance in this document have been developed to ensure a more uniform approach to the assessments of reference material producers, and the resulting scope of accreditation. The requirements are expressed in sentences in which the principal auxiliary verb is “shall” whilst recommendations are expressed in sentences in which the principal auxiliary verb is “should”.

Requirements and guidance given in this document are applicable to both certified and non-certified reference materials unless specifically mentioned the otherwise. Reference materials mentioned in the document mean both certified and non-certified reference materials.

AUTHORSHIP

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1. BACKGROUND

In October 2004, the ILAC General Assembly passed a resolution identifying the general approach that Accreditation Bodies (ABs) shall take when assessing and accrediting reference material producers (RMPs). Below are the ILAC GA resolutions relating to RMP accreditation.

ILAC Resolution GA 8.11

The General Assembly acknowledges that assessing the technical competence of bodies producing reference materials with assigned values is accreditation of a conformity assessment activity.

ILAC Resolution GA 8.12

The General Assembly resolves that accreditation of technically competent bodies producing reference materials with assigned values will be conducted against harmonized criteria based on ISO Guide 34 and ISO/IEC 17025 in combination.

ILAC Resolution GA 9.28

Following 2004 ILAC GA resolutions 8.11 and 8.12 relating to accreditation of Reference Materials Producers, the General Assembly resolves that the accreditation to ISO Guide 34 in combination with ISO/IEC 17025 be included under the current ILAC arrangement when appropriate procedures for this activity are developed and agreed by ILAC.

ILAC Resolution GA 16.20

As ISO Guide 34:2009 includes normative references to ISO/IEC 17025 and ISO 15189, the General Assembly resolves that accreditation of reference material producers is conducted in accordance with ISO Guide 34:2009 alone. This resolution rescinds ILAC resolution 8.12. In addition ILAC R6:07/2012 will be amended to reflect this decision.

In 2005, the APLAC MRA Council resolved (MRA RES 15.22) that “the inaugural signing of the extended APLAC MRA take place once 4 accreditation bodies have been evaluated and have met all the conditions for the extension of their scopes of recognition to include RM producers”.

The inaugural signing of the APLAC MRA for RMP accreditation took place on 7 December 2007 in Kuala Lumpur, Malaysia. The four inaugural signatories are A2LA, NATA, CNAS and IAJapan. A list of signatories to the APLAC MRA is obtainable from http://www.aplac.org/aplac_mra.html. It is acknowledged that there are other APLAC MRA signatories that are keen to implement RMP accreditation programs. However, the operations of RMPs can vary over time and in relation to the RMs produced. Questions have been raised about the appropriate ways to approach an assessment that presents such a complexity of activities, and also how to define the resulting scope of accreditation.

The APLAC MRA Council asked the APLAC Technical Committee to address the complexities and activities associated with the assessment and accreditation of a RMP. Issue 1 of this document was intended to serve that purpose. Subsequently, it was recognized that some guidance given in issue 1 of this document was essential for the harmonized and consistent application of the accreditation criteria. Therefore this document

was reviewed with a view to identify those points which should be considered as mandatory. It should, however, be stressed that no additional accreditation requirements are imposed on the RMPs. This document only serves as an application document for the accreditation criteria. It may, however, include some requirements for the assessment procedures, the presentation of the scope of accreditation and supplementary criteria arising from the APLAC MRA. To maintain its usefulness, guidance has been retained in the document.

The last edition of this document has taken into account the experience gained in the assessment and accreditation of RMPs by some APLAC members. The outcome from an APLAC workshop on the accreditation of RMPs, held in Tsukuba, Japan in 2007, has also been included. A conclusion that this workshop drew is that, depending on the actual situation of the RMP and the RMs it produced, general guidelines or principles for the application of ISO/IEC 17025 that are relevant to the activities of RMP tasks are more useful than specific guidelines. Specific guidelines may lead to rigidity and hamper the exercise of professional judgment. This might undermine the assessment of competence. Examples of the application are useful. Hence, it is not the intention of this document to give specific guidelines for the application of ISO/IEC 17025.

In 2009, a new edition of ISO Guide 34 was published. During the revision of ISO Guide 34, ISO REMCO considered the views and suggestions of APLAC, in particular, the outcome of the APLAC Workshops on Reference Material Producer Accreditation. One of the main drivers for the “major” revision of ISO Guide 34 was the strong endorsement by APLAC. APLAC also provided comments and inputs at each stage of the revision process. As a result, many of the requirements in the previous edition of APLAC TC008 have been taken on board by ISO REMCO in the 2009 edition of ISO Guide 34.

The main purposes of this revision are firstly to address ILAC GA 16.20 that no longer requires ISO Guide 34 and ISO/IEC 17025 in combination, and secondly to remove those items that have already been included in the 2009 edition of ISO Guide 34. In addition, guidance on the application of ISO Guide 34 to assessments and accreditation has been significantly expanded in light of the experience gained in the years since the publication of the last edition of this document. By providing further guidance, accreditation bodies developing RMP accreditation programme may find this document more useful. Further, this document can assist in future harmonization of RMP assessments and accreditation amongst APLAC members, as well as assist those accreditation bodies that wish to establish such a programme.

It should be stressed that, although the ILAC resolution directed the use of ISO Guide 34 to accredit RMPs, ISO Guides 30, 31 and 35 and the VIM are normative-references in ISO Guide 34. In fact, in the 2009 edition of ISO Guide 34, it has specifically mentioned that the RMP shall have a management system that covers the assignment of property values, including preparation of certificates or statements in accordance with ISO Guide 31 when appropriate and compliance with ISO Guide 30 and with appropriate sections of ISO Guides 35.

Also, the gap analysis between ISO Guide 34 and ISO/IEC 17025, and their cross referencing in the previous edition (i.e. Annexes 1 and 2), have been removed because the

2009 edition has already provided this cross-reference table in Annex C of ISO Guide 34: 2009.

The value of accreditation to ISO Guide 34 and the application of the accreditation requirements in ISO Guide 31 are to ensure that appropriate information is available in the certificates of any CRM purchased for the purpose of conducting method validation, ensuring metrological traceability of measurement results, and/or instrument calibration. Lack of this information will impact the accuracy and validity of test and calibration results. The APLAC MRA for accreditation of RMPs should enable regulators and users to purchase reference materials from competent and appropriately accredited RMP.

2. INTRODUCTION

The following principles apply to the assessment and accreditation of RMPs.

- 2.1 At a minimum, the RMP is the organization responsible for the project planning and management, assignment of and decision on property values and relevant uncertainties, authorization of property values and issue of the certificate or other statements for the reference materials (RM). It is the recognized legal entity that can be accredited. The RMP can be considered a “producer” but cannot be considered solely a “laboratory”. The production of RMs involves some activities that are not normally considered the activities of a laboratory. The term “production” is defined by ISO Guide 34:2009 under clause 3.3. When used in this document, it includes all necessary activities and tasks leading to a RM supplied to customers, and includes at least those listed above. Where an organisation only provides services such as the provision of reference values to a candidate RM, it shall not be considered as a RMP.
- 2.2 The RMP shall be assessed for the purpose of accreditation in accordance with ISO Guide 34. As there are many different approaches a RMP can use to generate and assign values to its reference materials, the relevance of a requirement should be assessed in the context of the tasks performed. Tasks that may be critical for one RMP may not be as important or may even be insignificant for another. RMs can range from the fairly simple e.g. calibration solution, to the very complex e.g. DNA sequence. The assessment team should perform an analysis of the RMP operation and determine how the accreditation criteria documents should be applied based on the principles, guidance and requirements given in this document.
- 2.3 The following table provides examples, numbered 1 to 8, of how tasks of RM production (identified below in Table 1) may be shared between the RMP and their subcontractor (where applicable). This table is offered for the purpose of description and should not be considered to provide an exhaustive coverage or to be all inclusive of every possible or probable RMP/subcontractor arrangement. The ISO document(s), i.e. ISO Guide 34 and the normative references, listed in the second column are considered to contain requirements that are relevant to the respective tasks listed in the first column.

Stages/ Tasks of RM production	ISO Documents	Example Scenarios and Responsible organisations							
		1	2	3	4	5	6	7	8
<i>Production planning</i>	ISO Guide 34 + ISO/IEC 17025	R	R	R	R	R	R	R	R
# Material preparation**	ISO Guide 34 + ISO/IEC 17025	R	S	S	S	S	R	S	R
# Homogeneity/ Stability testing	ISO Guide 34 + ISO/IEC 17025	R	R	R	S*	S*	S*	S*	R
# Characterization of Property Values	ISO Guide 34 + ISO/IEC 17025	R	R	R	S*	S*	S*	R	S*
<i>Assignment of and decision on Property Values</i>	ISO Guide 34 + ISO/IEC 17025	R	R	R	R	R	R	R	R
<i>Authorization of property values and issue of certificate</i>	ISO Guide 34	R	R	R	R	R	R	R	R
# Handling and storage (including post certification testing)	ISO Guide 34 + ISO/IEC 17025	R	R	S	R	S	S	S	R
Distribution & post distribution service	ISO Guide 34	R	R	S	R	S	R	S	R

Tasks denoted by *italics* shall be performed by the RMP

R = Tasks performed by the RMP

S = Task performed by subcontractor

If performed by a subcontractor, the RMP shall ensure the technical competence of that subcontractor

* Any conclusions in regards to these tasks shall be made by the RMP

** Testing, calibration and measurement activities involved in material production and preparation shall, as cross-referenced in ISO Guide 34, comply with the relevant parts of ISO/IEC 17025

2.4 The following are some existing modes of operation of an RMP.

- a) A single organisation produces the candidate RM and assigns the property values based on its own measurement results (example 1 as given in the above table).
- b) An organisation produces the candidate RM and assigns the property values based on the measurement results from other (subcontractor) laboratories. Handling and storage of the RM are performed by the subcontractor. The certificate is issued by the producer (example 6).
- c) An organisation produces the candidate RM and is responsible, as an example, for the homogeneity and stability studies. The property values are characterized by an external accredited or non-accredited laboratory. The producer makes a decision on the values, issues certificate and sells the RM (example 8).
- d) An organisation subcontracts the preparation of a candidate RM and then assigns the property values based on measurement results from its own laboratories. The organization that issues the certificate sells the RM (example 2).
- e) An organisation subcontracts the production of a candidate RM and all laboratory work necessary to assign the RM property values. The certificate is issued by the RMP and the RM is distributed by the RMP or an external party (example 5).

3. PERFORMING THE ASSESSMENT OF A REFERENCE MATERIAL PRODUCER

- 3.1 The accreditation body (AB) should, based on the modes of operation and the activities performed by the applicant RMP or a RMP seeking an extension in scope, determine the most applicable combination of accreditation criteria. ISO Guide 34 provides references to specific requirements given in other ISO Standard and Guides for various activities. These requirements given in the referenced ISO Standard and Guides also form parts of the accreditation criteria for RMPs.
- 3.2 An AB should also investigate available technical standards that might be included to support accreditation for sector specific RM programs to underpin the technical rigour of the discipline specific program (e.g. ISO 6141:Requirements for certificates for calibration gases and gas mixtures; ISO 6142:Gas analysis- Preparation of calibration gas mixtures – Gravimetric method.) The scope of accreditation should refer to these technical standards when they are used by the RMP.
- 3.3 It should be noted that a RMP may operate in different modes at different times for different RMs. This shall be taken into account in the assessment.
- 3.4 The AB shall obtain information from the RMP that demonstrates how the RMP is organised and how it meets the relevant requirements of ISO Guide 34 and other normative references. The information shall include but not be limited to the following.
- (i) The types or categories of RMs that are produced by the RMP.
 - (ii) The technical standards and/or specifications that are being used for the production of RMs.
 - (iii) The tasks/activities performed by the RMP itself in relation to the specific RMs.
 - (iv) The proposed scope of the RMP's use of subcontractors and the tasks/activities that are performed by these entities, and the RMP's register of all subcontractors used, which should include the following information;
 - name and address of the subcontractor,
 - the scope of tasks/activities performed by each subcontractor,
 - the testing, calibration and measurement activities performed by the subcontractors,
 - evidence of the technical/quality competence of the subcontractors (e.g. certification to ISO 9001 for non-testing activities and/or accreditation to ISO/IEC 17025 for testing, calibration and measurement activities).
 - (v) ISO/IEC 17025 accreditation for the testing/calibration performed internally by the RMP.
- 3.5 The applicability of ISO Guide 34 and ISO/IEC 17025 requirements to the operation should be considered on a case by case basis because they may vary with the

actual situation of the RMP and the RMs they produce, and they may also change as the production process changes. It should be emphasized that the RMP is accredited for producing specified RMs using the production procedures. The accreditation therefore covers all batches/lots of RMs within the scope of accreditation of the RMP subsequent to granting of the accreditation. It may however also include existing batches/lots that have been produced and demonstrated by the RMP as having met the accreditation criteria and in turn, has been assessed by the accreditation body.

- 3.6 Assessment with respect to the RMP's compliance with ISO/IEC 17025 requirements for testing/calibration/measurement determinations activities should be given special attention.

A RMP facility may be considered as an organization whose business is the production (and certification) of RMs where testing and calibration activities are undertaken. Even if all of testing and calibration activities are carried out by outside parties, an RMP would still perform some activities related to measurements such as "review of requests, tenders and contracts", "subcontracting of tests and calibrations", "selection of methods", "estimation of uncertainty of measurement", "control of data", "sampling", "handling of test and calibration items", etc.

- 3.7 All requirements of ISO/IEC 17025 shall be applicable to testing and calibration activities of the RMP. Testing and calibration activities include not just performance of tests, calibrations, measurements, and sampling, but also other related activities such as review of requests, tenders and contracts, subcontracting of tests and calibrations, selection of methods, validation of methods, measurement traceability, assuring the quality of test and calibration results, etc.
- 3.8 Where the RMP is accredited to ISO/IEC 17025 by an APLAC and/or ILAC MRA signatory for the applicable testing/calibration activities, this accreditation can be considered satisfactory to meet the ISO/IEC 17025 requirements for testing or calibration competence. For subcontractors for testing and calibration, see clause 4.5.2 of this document.
- 3.9 If the RMP is not accredited to ISO/IEC 17025, but performs the testing or calibration activities, the AB's assessment of the RMP shall include an assessment of the RMP's testing and calibration activities against the relevant parts of ISO/IEC 17025.
- 3.10 The AB shall establish an appropriate assessment team and generate an appropriate agenda to cover the full breadth of the RMP's activities. The assessment team shall include competent technical assessors who have relevant knowledge of the categories of RMs produced and to cover the technical elements of an assessment based on ISO/IEC 17025 in addition to ISO Guide 34. The assessment team shall also include expertise in statistical methods and ISO Guide 35.
- 3.11 Because the RMP has overall responsibility for the competent performance of all RMP tasks, and for ensuring the suitability of assigned RM values and resulting RM certificates/statements, the AB's assessors shall give the following activities covered

by ISO/IEC 17025 particular attention during the assessment of the RMP, whether they are performed in house or by a subcontractor:

- a) Selection of test or calibration methods should be done in conjunction with sections 5.9, 5.15 and 5.16 of ISO Guide 34:2009. The degree of rigour of the RMP's processes for analytical method development and validation should be strong, and the assessors should include a robust assessment of metrological traceability and measurement uncertainty estimation, in line with the scrutiny given to a calibration laboratory.
- b) Measurement uncertainty estimations for the testing and calibration results shall be appropriately rigorous and properly verified, because these estimates will affect the RM assigned values.
- c) The suitability of the metrological traceability utilized by the RMP is important. In cases where the metrological traceability cannot be achieved through an unbroken chain of calibrations, clause 5.12.2 of ISO Guide 34:2009 provides other alternative means. If a CRM is used for establishing metrological traceability, the CRM used shall have comparatively small uncertainty (refer to Note below) and higher in the metrological traceability hierarchy. The uncertainties in the certified values of the CRM used shall be suitable for establishing metrological traceability appropriate to the RMs being produced.

Note: The RMP should consider the competence of the producer of any certified reference material it uses to provide the metrological traceability of the assigned value of its CRM. A competent RMP or testing/calibration organization which may be a National Metrology Institute which is a signatory to the CIPM MRA, participates regularly in BIPM or Regional Key Comparisons, and has the relevant CMCs been included in Appendix C of the BIPM Key Comparison Database (KCDB).

- d) Proficiency testing can be used to monitor the on-going competence of the testing and calibration processes. If the RMP performs testing and calibration that significantly affects the uncertainty of the assigned property value of a RM, the RMP shall participate in the proficiency testing programs as required in ILAC P9 for the tests and calibrations it performs. If a laboratory acts as a subcontractor, the RMP shall require the laboratory to participate in proficiency testing programs, as required to meet ILAC P9 for the tests and calibrations it performs. When proficiency testing programs are not available, other means to demonstrate competence, e.g. use of measurement audits and check samples, shall be considered.
- e) Clause 4.5.3 of ISO Guide 34:2009 requires the specifications for subcontractors executing measurements or testing to include requirements as described in ISO/IEC 17025. Though accreditation to ISO/IEC 17025 for testing and calibration activities is the most direct way to demonstrate technical competence, it is not mandatory and sometimes accreditation is not relevant or practical, e.g. when it is a very specialised and non-routine measurement, or the subcontractor

is a research facility that is used infrequently, or the subcontractor is used only once. In such cases, the RMP shall monitor the competence of these subcontractors in other ways. In addition to the ones given in ISO Guide 34:2009 clause 4.5.3 other ways such as provision of a method protocol with a rigorous quality control regime, and/or through proficiency testing (also see 3.11 d) above) may be used. Where there are difficulties in ensuring the competence of subcontractors, the RMP may consider using other means of determination of property values, e.g. by a collaborative study involving multiple laboratories (see clause 9.4.2 of ISO Guide 35:2006). Assessment of the competence of subcontractors for testing and measurement by purely requesting them to complete a questionnaire (i.e. self-declaration by the subcontractors) without on-site verification may not be sufficient. Other evidence to substantiate the claims should be obtained, e.g. reports of the PT schemes the subcontractor has participated, recent quality control data, copy of calibration certificates or certificates of CRMs used for calibration of equipment to demonstrate how the subcontractor establishes metrological traceability, etc. Further guidance/requirements are given in Sections below.

- 3.12 It should be emphasized that accreditation is granted to the RMP, and not to its subcontractors. In this regard, the RMP shall have a written agreement with its subcontractors. The agreement shall address the following issues. It is an obligation of an accredited RMP to ensure that its subcontractors follow and meet the applicable technical and, where relevant, non-technical requirements arising from the RMP's accreditation. Documentation issued to subcontractors as a result of a successful assessment by the RMP shall state that it is only for the purpose of the contract and is not certification nor accreditation. The AB may impose suspension of the accreditation when the RMP fails to ensure that its subcontractors follow any of its accreditation regulation or requirement relevant competency requirements for the subcontracted work.
- 3.13 The AB shall ensure that the assessment process establishes how the RMP determines the competence of its subcontractors to undertake the tasks they have been allocated in the production of each type of RM. For critical activities, the AB should, where necessary, witness a selection of examples of how the RMP evaluates the competence of its subcontractors on-site. This will be necessary where the competence of the subcontractor involved in the generation of measurement data for the characterisation of property values cannot be determined through the information provided by the RMP. It may also be necessary for the generation of other measurement data used by the RMP (such as for homogeneity and/or stability testing), depending on the criticality of the data to the integrity of the RMs produced. The accreditation body shall establish its policy whether it permits serial subcontracting (i.e. subcontracting of subcontracted work), on an on-going basis, by the RMP's subcontractors. It may be difficult in controlling the competence of the subcontractor who actually performs the work if serial subcontracting is allowed. Accordingly, the AB shall assess the means by which the RMPs ensure that the tasks performed by the final subcontractors comply with the specifications set by it for such tasks, as required by clauses 4.5.1 and 5.3.1 of ISO Guide 34:2009, should

be assessed. Further requirements and guidance for subcontractors are given in Section 4.5 of this document.

- 3.14 An on-site assessment of the subcontractor by the RMP is not normally needed if the testing or calibration subcontractor is accredited for the specific test or calibration, and the degree of the review done by the RMP of the contract with its subcontractor is appropriate and at a minimum includes a review of the:
- a) measurand required,
 - b) test and/or calibration method(s) used,
 - c) required measurement uncertainty,
 - d) metrological traceability,
 - e) reporting requirements,
 - f) performance of proficiency testing activities (where suitable and applicable); and
 - g) attention by the subcontractor to performing the work with the required technical rigour.

Note: The contracting referred to above is between the RMP and its subcontractors, and not between the RMP and its customers. As such, it relates to clauses 4.5 and 5.3, in addition to clause 4.4.3, of ISO Guide 34:2009.

- 3.15 When the testing or calibration is performed by the RMP, clause 4.1.4 of ISO/IEC 17025:2005 that covers conflict of interest shall be applied when assessing the in-house testing or calibration function and its relationship with the production department.
- 3.16 Sampling, sub sampling and sample size taken for production and testing are critical in the assignment of property values as well as homogeneity and stability testing. These shall be carefully assessed by the AB's assessors.
- 3.17 The resulting RM certificates/statements shall be rigorously assessed by the assessment team. The scope of the RMP is to produce RMs and certificates shall meet clause 5.17 of ISO Guide 34 and the requirements of ISO Guide 31.
- 3.18 ISO Guide 34 requires RM certificates to meet all of the *requirements* of ISO Guide 31, noting that ISO Guide 31 has many “should” statements that are not requirements; however these “shoulds” may be considered on a case by case basis as additional requirements where they are fundamental to the application and technical validity of the RM e.g. chemical, biological, etc.
- 3.19 If the AB is not satisfied with the RMP evaluation of the competence of its subcontractors (especially non-accredited testing or calibration subcontractors), the AB should arrange a visit to those subcontractors or should deny accreditation of the RMP until the RMP can provide further objective evidence of a satisfactory evaluation of its subcontractors.

Examples of unsatisfactory findings may include, but not be limited to:

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- a. The RMP did not do a second party audit of the subcontractor;
 - b. The RMP audit did not cover the subcontractor's tasks;
 - c. The subcontractor did not respond to the RMP's findings and/or the RMP did not follow up on the audit finding resolutions;
 - d. Non-conformities found by the AB's assessment team during the assessment support the concern that an adequate competency examination of the subcontractor was not done; and/or
 - e. No participation or inappropriate participation in PT activities or unsatisfactory investigation of outlying results from PT by the subcontractors.
- 3.20 The AB shall require that it is informed by the accredited RMPs, without delay, of any changes in the arrangements between the RMP and its subcontractors. Upon such notification the AB shall properly assess the new subcontractor arrangements in order to determine whether accreditation requirements continue to be met.

4. GUIDANCE ON THE APPLICATION OF SECTION 4 OF ISO GUIDE 34:2009

4.1 Management system requirements

4.1.1 The management system of a RMP need not be complex and its format will depend on a number of factors including the size of the RMP, number of staff members and the range, volume and complexity of the work it performs. In cases where a RMP is part of a larger organisation, RMP activities may already be incorporated in a document covering the organisation's total range of operations.

4.1.2 Clause 4.1.2 c) requires the RMP to conduct all testing and calibration in support of the production of reference materials in compliance with the requirements of ISO/IEC 17025. Guidance on the assessment of compliance is given in clauses 3.7, 3.8 and 3.9 of this document.

4.2 Organisation and management

4.2.1 The technical management may be a designated technical manager or may consist of a combination of designated technical managerial personnel each of them responsible for specified areas. The responsibility of technical issues for all accredited activities shall be fully covered by the technical management.

4.3 Document and information control

No additional application notes.

4.4 Request, tender and contract reviews

4.4.1 When reviewing requests, tenders and contracts, RMPs shall ensure that the requested matrix, property values and their metrological traceability and measurement uncertainty meet the need of the customer. In some cases, the stability time required should also be included in the review. If necessary, the RMP should give advice to the customers and help them to determine their needs.

4.5 Use of subcontractors

4.5.1 RMPs shall document, in the quality manual or related documents, their policy and procedures for sub-contracting. There are some processes that are not allowed to be subcontracted. Such processes are listed in clause 5.3.1 of ISO Guide 34: 2009 (see also NOTE 1 under clause 3.2 of ISO Guide 34:2009).

4.5.2 For testing and calibration subcontractors, ISO Guide 34 accepts accreditation to ISO/IEC 17025 as evidence of their competence. The accreditations should be granted by accreditation bodies which are signatories to the respective ILAC MRA and the scope of accreditation should cover the specific tests and/or calibrations performed in support of producing the RM.

4.6 Procurement of services and supplies

4.6.1 There are two commonly encountered situations where a RMP will have to procure services and supplies:

- (a) Procurement of consumables or perishable items, e.g. media, chemical reagents and glassware:

Records should be kept of the different brands of those items which bear a critical influence on the quality and validity of the RMs. The record should where appropriate include results of the acceptance tests on each new batch prior to use. Consideration should be given to excluding supplies from suppliers that are inappropriate/invalid or have a known history of supplying inappropriate/invalid materials.

- (b) Procurement of equipment:

It is further recommended that when choosing suppliers of service or products, priority should be given to 3rd party certified (refer to IAF Multilateral Mutual Recognition Arrangement) and accredited (refer to ILAC Mutual Recognition Arrangement) organizations.

When purchasing of candidate material for further processing, requirements given in other sections such as clauses 5.4, 5.5, 5.8 etc. of ISO Guide 34:2009, should be followed. Regarding testing and calibration service, the provisions on use of subcontractors for testing and calibration apply.

4.7 Customer service

4.7.1 There are requirements for customer service given in other sections of ISO Guide 34 , e.g. clauses 5.14.5 and 5.18.4, etc.

4.8 Complaints

No additional application notes.

4.9 Control of non-conforming work and/or reference materials

4.9.1 Common examples of non-conforming work include environmental conditions in the testing or calibration areas exceeded the specified limits, tests performed using instruments with overdue calibration, acceptance criteria of quality control not met, unsatisfactory performance in proficiency testing schemes, etc. It is important that RMPs should not just correct the problem but shall initiate actions according to the requirements given in clause 4.9.1 of ISO Guide 34:2009, which include a determination the significance of the non-conforming work and this should include an investigation of whether the non-conforming work is an isolated incident or is due to some underlying causes with a possibility of recurrence. In the latter case, requirement given in clause 4.9.2 of ISO Guide 34:2009 should be conformed to. This implies that corrective actions, in addition to corrections, are also needed. It should be emphasized that all personnel of the RMP need to be familiar with the

procedures for handling non-conforming work and/or reference materials. They should follow the documented procedures whenever non-conforming work and/or reference material is identified. Training on the procedures is essential to ensure relevant staff understands the procedures.

- 4.9.2 Records of nonconforming work and/or reference materials should be maintained as part of the RMP's quality records (see clause 4.13.1.1 a and b) of ISO Guide 34:2009). The records should include information on the nonconforming work and/or reference materials, actions taken, results of evaluation of the significance and extent of the nonconforming work, etc. Internal audit should include checking the effectiveness of implementation of this aspect.

4.10 Corrective actions

No additional application notes.

4.11 Preventive actions

No additional application notes.

4.12 Improvement

No additional application notes.

4.13 Records

- 4.13.1 Technical records shall, as applicable, include all original observations and raw data and provide a traceable link between the reference materials produced and the information on the certificates or documentation of the reference materials. This applies equally to electronic and paper record systems. If a RMP uses an Information Management System, the system should meet all the relevant requirements, including audit trail, data security, safety and integrity, etc. It should be fully validated and records of validation should be maintained. RMPs should keep back-up copies of electronic records within their retention period. They should also have a system to ensure that electronic records remain accessible within that period even though the hardware and software of their computer system are being updated from time to time.

- 4.13.2 The record system should allow for ready retrieval of original observations and data pertinent to any issued reports or certificates.

- 4.13.3 For each reference material produced, the records system should retain and provide ready access to the following detailed information:

- (i) the full description of the reference material;
- (ii) the unique identification of the reference material;

- (iii) the test or calibration method or procedure used in the production process;
 - (iv) identification of equipment and reference materials used in the production process;
 - (v) all data relating to the preparation and manufacturing of the candidate materials;
 - (vi) original observations during the test or calibration and calculations based on the observed data;
 - (vii) data and the statistical calculations for homogeneity and stability studies;
 - (viii) data used in the assignment of property values and their uncertainties, including those data which have been rejected and the reasons for rejection;
 - (ix) identification of persons performing the work;
 - (x) an exact copy of the issued documentation or certificate of the reference material produced.
- 4.13.4 Original observations should be recorded immediately into bound notebooks, or onto properly designed proforma worksheets using indelible pen. Instrument printouts should be kept when they are available. Where data processing systems are used, records of raw data should be retained unless data are automated.
- 4.13.5 Errors in calculations and incorrect transfer of data are major causes of incorrect results. Calculations and data transfers should be checked by another person, then initialed and dated by the reviewer except in the case when there is no other suitable person available for this purpose.
- 4.14 Internal audits
- 4.14.1 Internal auditors shall be familiar with the requirements of ISO Guide 34 and the requirements of ISO Guides 30, 31, 35 and ISO/IEC 17025 (or ISO 15189 for medical reference materials), and the requirements of this document.
- 4.15 Management reviews
- No additional application notes.

5. GUIDANCE ON THE APPLICATION OF SECTION 5 OF ISO GUIDE 34:2009

5.1 General

- 5.1.1 There are technical standards published by international organisations, or other well recognised professional bodies, that may be applicable to the production of certain reference materials. If such applicable technical standard exists, a RMP should, as far as possible, follow such standards in its production of the reference materials (see section 3.2 of this document and clause 5.4.1 of ISO Guide 34:2009).

5.2 Personnel

No additional application notes.

5.3 Subcontractors

Application notes are given under Section 4.5 of this document.

5.4 Production planning

- 5.4.1 It is critical that, before the start of the production of reference material, a detailed production plan is available. It is understood that pilot studies may sometimes need to be carried out but the need of any pilot study should be considered at the planning stage. The production plan should be fully documented. There are requirements for each step of the production process given in ISO Guide 34 and the RMP is required to provide evidence that, at the planning stage, these requirements are given full consideration, and if necessary, recommendations from advisory groups have been sought.

- 5.4.2 The production plan may need to be reviewed regularly during the production process. If it is necessary to make any change to the plan, the effects of the change on the conformity with the requirements of ISO Guide 34 should be evaluated. Changes should be approved by the person authorised, in accordance with clause 5.2.6 of ISO Guide 34:2009, to perform production planning of the reference material. Changes should be fully documented, and should include the reasons and justifications for the changes. If the changes can affect the contract with the customer, the customer should be consulted. Customer's agreement with the changes should be obtained and records maintained as required by clause 4.4.2 of ISO Guide 34:2009.

5.5 Production control

- 5.5.1 Although effective control of each stage of the production process is needed, there are also certain critical steps in each stage where the quality of the reference material can be significantly affected. An analysis of such critical control points can be carried out and a plan that is designed to ensure that these critical control points are effectively controlled, and monitored, is a useful means to ensure the quality of reference materials.

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- 5.5.2 Records shall be maintained to provide evidence that there is effective control of each stage of reference material production, e.g. records of inspection, testing, etc
- 5.6 Accommodation and environmental conditions
- 5.6.1 Suitability of the accommodation and environmental conditions for the production of a specific reference material should be assessed based on their effect on the quality and validity of the reference material being produced, including how they affect the:
- (a) integrity of the reference materials;
 - (b) performance of laboratory equipment;
 - (c) competent performance of laboratory staff;
 - (d) compliance with the conditions specified in the production plan.
- 5.6.2 Consideration of environmental effects on reference materials includes precautions necessary to prevent contamination and degradation (refer to 5.7.2 of ISO Guide 34). The areas for the material preparation, preconditioning, testing or calibration and storage should be of adequate size, free from dust, fumes and other factors (such as excessive temperature, humidity and direct sunlight) which may affect the integrity of the reference materials. If the reference materials produced require refrigeration, refrigerators or freezers of adequate capacity and capable of maintaining the required temperatures, these shall be available and temperature monitored.
- 5.6.3 The potential effects of environment on equipment performance include corrosion, temperature, humidity, vibration, electrical power stability, dust and electromagnetic influences. The location of all items of equipment likely to be affected by these factors should be chosen to eliminate or minimise any adverse effects.
- 5.6.4 Accommodation and environmental conditions should also be assessed based on their effects on staff competence in performing specific activities. There should be sufficient space available for staff to perform their duties comfortably, with adequate provision of lighting and with precautions taken to minimise noise.
- 5.7 Material handling and storage
- 5.7.1 It should be emphasized that the requirements of this section apply to all stages of the production - from the receipt of the raw material to the finished reference material. If during some stages of production, the material has to go out of the direct control of the RMP, the RMP should provide necessary instructions to the party responsible for handling the material. When storing the material, the storage environmental conditions should be specified.
- 5.7.2 When the same equipment is used for different materials, the facility should ensure that no cross-contamination or carry-over contamination is taking place.
- 5.7.3 All persons handling the materials should be trained on the proper handling procedures. They should be aware of the precautions to be taken whilst handling the material, as required by clause 5.2.2 of ISO Guide 34:2009.

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- 5.7.4 It is the responsibility of the RMP to ensure that the packing and labelling of the reference materials meet the safety and transport regulatory requirements. However, it should be emphasized that assessment of the compliance with the regulatory requirements is outside the authority of accreditation bodies.
- 5.8 Material processing
- 5.8.1 When the same equipment is used for processing different materials, the equipment should be thoroughly cleaned between uses to prevent possible cross-contamination.
- 5.8.2 All material processing procedures should be carried out by trained personnel and requirements of clause 5.2.2 of ISO Guide 34:2009 are applicable.
- 5.8.3 Preparation of the material (such as drying, mixing of ingredients, spiking with analytes, etc) is a form of material process.
- 5.8.4 When candidate reference materials are sent to subcontractors for testing, they should be uniquely labelled, suitably packed and stored in suitable conditions during transport. Instructions on the storage conditions should be given to the subcontractors.
- 5.8.5 In cases where the certified values are based on data obtained in the material processing procedure, the requirements relating to the assignment of property values and their uncertainties apply to the material process procedures. In such cases, the material process procedures should comply with the requirements for measurement methods and metrological traceability given in Sections 5.9 and 5.12 of ISO Guide 34:2009. The requirements for measuring equipment given in Section 5.10 of ISO Guide 34:2009 also apply to those items of equipment use in the material processing stage which contributes to the uncertainty of the assigned values of the reference materials.
- 5.9 Measurement methods
- No additional application notes.
- 5.10 Measuring equipment
- No additional application notes.
- 5.11 Data evaluation
- 5.11.1 Homogeneity and stability assessments, characterisation and assignment of property values and their uncertainties all involve evaluation of data. The RMP shall use appropriate statistical techniques for data evaluation. The general and statistical principles for certification of a given reference material in ISO Guide 35, where appropriate, shall be followed.

5.12 Metrological traceability

5.12.1 See clause 3.11 c) of this document.

5.13 Assessment of homogeneity

5.13.1 Under normal circumstances, the degree of homogeneity assessment of a RM with respect to the property of interest should be performed. It is not acceptable to assume the homogeneity of a property value based on the assessment of another value unless correlation is demonstrated with analytes that are tested for homogeneity. See the requirement given in clause 5.12.3.1 of ISO Guide 34:2009 in case homogeneity testing is done only on a subset of the assigned values.

5.13.2 When data from assessment of homogeneity are used for assigning the property values, the requirements for metrological traceability (Section 5.12 of ISO Guide 34:2009) and characterisation (Section 5.15 of ISO Guide 34:2009) apply to the test procedures used.

5.14 Assessment of stability

5.14.1 Under normal circumstances, stability assessment for each and every certified property value should be performed. It is not acceptable to assume the stability of a property value based on the assessment of another value unless correlation is demonstrated with analytes that are tested for stability.

5.14.2 Prediction of stability using a model is generally not acceptable unless such model is well established and widely accepted in the discipline concerned.

5.14.3 In cases where data from assessment of stability are used for assigning the property values, the requirements for metrological traceability (Section 5.12 of ISO Guide 34:2009) and characterisation (Section 5.15 of ISO Guide 34:2009) apply to the test procedures used.

5.14.4 Stability assessment should include assessment of the effects of shipment. This includes studies with actual shipping under maximum stress conditions, e.g., distance, and temperature.

5.14.5 Stability assessment should include assessment of the effects of use. This includes studies with multiple subsamples and any requirements for changed temperature for storage before subsampling. Any associated uncertainty could be expressed within the long term stability assessment or as considerations described in the certificate.

5.15 Characterisation

5.15.1 When a property value of interest is derived from an “empirical method”, the RMP should use that particular “empirical method” for characterisation. Details of the characterisation procedures used should be recorded. When more than one laboratory is engaged for characterisation, then all of them should use the same

“empirical method”. Such property values are only meaningful when applied to the same “empirical method”. Therefore, to be more useful, the empirical methods used should be those published by standard writing bodies or widely recognised professional bodies in the field concerned.

5.16 Assignment of property values and their uncertainties

5.16.1 As CRMs are often used by laboratories for establishing their metrological traceability, it is important that the uncertainties of the assigned values are estimated using methods which are generally more rigorous than for other purposes. The uncertainties include not just the measurement uncertainty of the characterisation procedure but also other contributions.

5.16.2 Uncertainty in this Section covers both “measurement uncertainty” of a quantity value and “uncertainty” associated with a nominal property (i.e. property of a phenomenon, body, or substance, where the property has no magnitude e.g. colour chart, DNA sequence, etc).

5.16.3 The estimate of uncertainty should include at least the effects of characterization, homogeneity, transport, and long term storage. If any of these effects are known to be zero, that can be described here.

5.17 Certificates or documentation for users

5.17.1 A RMP may perform tests or calibrations for the production of reference materials. Such tests or calibration should be performed in accordance with relevant requirements of ISO/IEC 17025. It is however be noted that the requirement for reporting of results (i.e. clause 5.10 of ISO/IEC 17025:2005) only applies to internal testing and calibration reports and does not apply to certificates and documentation of the reference materials issued to users. See also clause 4.13.2 of ISO Guide 34:2009.

5.17.2 A RMP is allowed to contract out some of its tasks to competent subcontractors. It may not be necessary to indicate which parts of the production process have been subcontracted in the certificate of CRMs or the documentation for RMs.

5.17.3 Certificates or documentation for a certified reference material or non-certified reference material should contain a unique identification of its production process. This identification may take the form of a reference number, the name of the process or in other suitable information.

5.18 Distribution service

No additional application notes.

6. DESCRIBING THE SCOPE OF ACCREDITATION FOR A REFERENCE MATERIAL PRODUCER

- 6.1 An AB shall provide a scope of accreditation that describes the specific types of RMs that the RMP is competent to produce. Although RMP accreditation conveys competence as a producer, not as a laboratory, testing and/or calibration are integral components of RM production. It is permissible by ISO Guide 34 for a RMP to contract out certain tasks to competent subcontractors because the RMP itself lacks the necessary competence for the tasks. As such, the competence of a RMP to produce a RM depends on how that particular RM is being produced. A task which is originally performed by a competent subcontractor cannot be carried out by the RMP itself unless its competence in that task has been demonstrated to its AB. For example, characterisation of a RM by a single (primary) method may be carried out by a competent laboratory but may not be by the RMP itself if it does not have the expertise to enable it to ensure metrological traceability (see clause 5.15 of ISO Guide 34:2009). It may also be possible that the RMP may lack the necessary equipment for the tasks (e.g. homogenizer for homogenizing the candidate material, measuring equipment for characterisation, etc.). Consequently, the scope of accreditation, and/or assessment records/reports, if necessary, shall describe the specific activities and tasks, including those tests, calibrations and measurements, and possibly those tasks given in Table 1 for which the RMP is accredited. For some types of RMs as well as for certain property values, a RMP may only be competent to produce a particular range and within a certain uncertainty of the property value. In such case, the scope shall include the range of property values for the RMs with the associated uncertainties, where relevant, for which the RMP is accredited. It is also recognized that the property values and the associated uncertainties for certain RMs may vary between batches/lots of RMs although they are produced by the same accredited production procedures. These variations should however be within the accredited ranges and uncertainties in order for them to be considered as covered under the scope of accreditation.
- 6.2 Categories and sub-categories of reference materials are given in Appendix B of ILAC G12 and this appendix can serve as good guidance to describe the specific types of RMs that a RMP is accredited to produce. Sub-categories may be required and can be included for clarity in the RMP's scope of accreditation. This information could be useful to the customers of the RMP.
- 6.3 The Certificate of Accreditation shall state that the RMP meets the requirements of ISO Guide 34 and may also include a suitable reference to the relevant requirements of ISO/IEC 17025. The certificate of accreditation awarded to the RMP shall not imply this accreditation allows the RMP to operate as an accredited test/calibration laboratory (although separate laboratory accreditation may be held by the RMP).
- 6.4 If the RMP requests accreditation as a laboratory to ISO/IEC 17025 for its testing, calibration or measurement activities, this accreditation can be expressed in a separate scope and certificate of accreditation. In this case, all the criteria for laboratory accreditation apply.

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- 6.5 As an RMP can do various tasks (refer to Table 1), accreditation shall be granted to it for those activities that has been assessed and found to meet the relevant requirements. The scope of accreditation, or other records/reports that support the scope, shall clearly state these activities, together with the RM(s) that the RMP is accredited to produce. If a RMP does certain activities that are outside the scope of its accreditation, it shall not claim that it is accredited for producing the RM concerned, and cannot use an endorsed certificate/statement for such a RM.

ANNEX 1: Application of ISO/IEC 17025 to the RMP Tasks
(Informative)

Stages/Tasks of C(RM) production	Relevant ISO/IEC 17025 Clauses
<u>Production planning</u>	
> Design project	Not applicable
> Select collaborators (if needed)	4.4.3 (contract review); 4.5 (subcontracting)
Material preparation	
> Production planning	Not applicable
> Selection of materials	4.6 (purchasing)
> Production control	Not applicable
> Control of environment	5.3 (Accom. & environment.)
> Control of poor quality material	4.9 (Nonconforming product)
Homogeneity/Stability testing	
> Statistical Plan (ISO Guide 35)	5.3 (accom & envir.); 5.4 (test method & validation); 5.4.6 (MU); 5.5 (equipment); 5.6 (traceability); 5.7 (sampling)
> Calibration/testing homogeneity & stability (pre & post certification)	5.7 (sampling); 5.3 (accom. & environment); 5.4 (test method & validation); 5.5 (equipment)
<u>Characterization of Property Values</u>	
> Characterization of measurand	4.4 (contract review); 4.5 (subcontracting); 4.13 (records); 5.2 (personnel); 5.3 (accom. & environment); 5.4 (test method & validation); 5.5 (equipment); 5.6 (traceability); 5.7 (sampling); 5.8 (handling items); 5.9 (QC); part of 5.10 (reports)
> Method selection	5.4 (test method & validation)

Stages/Tasks of C(RM) production	Relevant ISO/IEC 17025 Clauses
> Method validation	5.4 (test method & validation)
> Quality assurance	5.9 (QC)
> Traceability of measurement results	5.5 (equipment); 5.6 (traceability)
> Uncertainty of measurement	5.4.6 (MU)
<i>Assignment of and Decision on Property Values</i>	
> Data evaluation	5.4.6 (MU); 5.4.7 (data control)
> Identification of bias	5.9.2 (QC)
> Statistical procedures	5.4.1 (test method & validation)
> Traceability of property values to SI/stated reference	5.6 (traceability)
> Uncertainty of property value	5.4.6 (MU)
<i>Authorization of property values and issue of certificate</i>	Not applicable
<i>Handling and storage (including post certification testing)</i>	
> Packaging & labeling	5.8 (handling items)
> Monitoring integrity of RM during storage	5.8 (handling items)
> Monitoring storage temperature & environment	5.3 (accom. & environment.) (4.13: records)

Note: The above are just examples of some of the ISO/IEC17025 requirements that are considered as applicable, in most of the cases, to the RMP tasks.